

Multispectral Analysis GUI (msaGUI) User Guide

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PREFACE

msaGUI enables user-friendly batch-analysis operations and visualization of multichannel (e.g. multispectral) data. Code repository and bug-reporting available on Github:

<https://github.com/cdavislab/multispectral-analysis>.

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1. Installation

1.1 Windows OS

Download the msaGUI folder directly from <https://github.com/cdavislab/multispectral-analysis> via **Releases**. On the releases page, download msaGUI_Windows.zip. Once downloaded, unzip the folder if needed.

Open msaGUI using the executable for your operating system: msaGUI.exe. Opening the file may take several seconds.

The code was tested on Windows 11 and does not require you to download and install Python. If your Windows version is not supported, the code can be run via its Python source code. Follow the instructions in 1.3 Source Code.

1.2 Mac OS

Download the msaGUI folder directly from <https://github.com/cdavislab/multispectral-analysis> via **Releases**. On the releases page, download msaGUI_Mac.zip. Once downloaded, unzip the folder if needed.

Open msaGUI using the executable for your operating system: msaGUI_Mac. If the file is zipped (i.e. the file is msaGUI_Mac.zip), first unzip the file, then open the application. Opening the file may take several seconds.

The code was tested on Ventura 13.0 and does not require you to download and install Python. If your Mac version is not supported, the code can be run via its Python source code. Follow the instructions in 1.3 Source Code.

1.3 Source Code

If your operating system, Windows version, or Mac version is not supported, the code can be run via its Python source code. The code was tested on Python 3.11.5.

1. Install Python3 from <https://www.python.org/downloads/>.
2. Download the msaGUI folder directly from <https://github.com/cdavislab/multispectral-analysis> via Code > Download ZIP. Once downloaded, unzip the folder if needed.
3. Install the requisite Python dependencies as listed on the project's Github in requirements.txt.
 - a. Open Command Prompt on Windows or Terminal on Mac/Linux.
 - b. Change to the directory of the files downloaded in instruction 2. Specific navigation will depend on the location that the folder is saved. For example:
on Windows: `cd C:\Users\Name\Downloads\msa_gui`
on Mac/Linux: `cd /Users/Name/Downloads/msa_gui`
 - c. Install the required Python dependencies
on Windows: `pip install -r requirements.txt`
on Mac/Linux: `python3 -m pip install -r requirements.txt`
4. Run the msa.py code in the command prompt/terminal window:
on Windows `python msa.py`
on Mac/Linux `python3 msa.py`

2. Tutorial: Running msaGUI

Open msaGUI for your operating system as described in **1. Installation**:

on Windows OS run the executable: msaGUI_Windows

on Mac OS run the executable: msaGUI_Mac

on Other OS or unsupported Windows/Mac versions open the command prompt or terminal window:

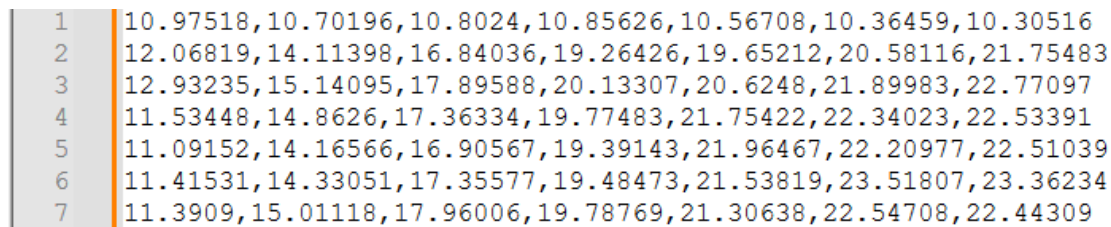
navigate to the folder with msa.py following instructions in **1.3 Installation part 3b**

run msa.py code on Windows python msa.py

run msa.py code on Max/Linux python3 msa.py

2.1 Data Preparation

Input files must be in the comma separated (CSV), tab separated (TSV), or TIFF format. TIFF files must not contain a stack of images. CSV files contain values of each pixel of an image with each column and row pair representing the pixel's value. Each pixel value is separated by a comma. For example, a 30 x 50 pixel image should be represented by 30 rows and 50 columns. TSV files are similar but separated by tabs. TIFF files are not stored as human-readable format. CSVs and TSVs should not include headers nor NaNs but can include negative values.



```
1 10.97518,10.70196,10.8024,10.85626,10.56708,10.36459,10.30516
2 12.06819,14.11398,16.84036,19.26426,19.65212,20.58116,21.75483
3 12.93235,15.14095,17.89588,20.13307,20.6248,21.89983,22.77097
4 11.53448,14.8626,17.36334,19.77483,21.75422,22.34023,22.53391
5 11.09152,14.16566,16.90567,19.39143,21.96467,22.20977,22.51039
6 11.41531,14.33051,17.35577,19.48473,21.53819,23.51807,23.36234
7 11.3909,15.01118,17.96006,19.78769,21.30638,22.54708,22.44309
```

Figure 1. Example of how raw data is stored in a csv file.

To perform a batch analysis, the program automatically groups files based on patterns in the filenames and user-provided keywords that denote the content of the file. Below are two valid grouping examples where the keywords are Signal1 and Signal2. The program looks for any instance of “Signal1” and “Signal2” in the file path. For each group, the file path must be exactly the same, except for the keyword. Files that do not match to full group will remain ungrouped and unprocessed.

Example 1 – Grouped based on folder name

Group 1: .../Sample1/Signal1_01-06-2026, .../Sample1/Signal2_01-06-2026

Group 2: .../Sample2/Signal1_01-06-2026, .../Sample2/Signal2_01-06-2026

Example 2 – Grouped based on file name

Group 1: .../Data/Sample1_Signal1, .../Data/Sample1_Signal2

Group 2: .../Data/Sample2_Signal1, .../Data/Sample2_Signal2

Example 3 – Grouping will not work because of ambiguity

Not Grouped: .../Signal1/Sample1_Signal1, .../Signal1/Sample1_Signal2

For this tutorial we have prepared six images that can be downloaded from the Github (see Preface):

DataFile_24hr_1655.csv

DataFile_24hr_1703.csv

DataFile_24hr_1744.csv

DataFile_72hr_1655.csv

DataFile_72hr_1703.csv

DataFile_72hr_1744.csv

2.2 Loading Data

To load images into the msaGUI:

1. Click the "Add" button in msaGUI
2. Navigate to the location that the images to be analyzed are saved. They should be prepared according to 2.1 Data Preparation. Select the file(s) to upload and click "open". (Note: multiple contiguous files can be selected by clicking the first file, holding down the Shift key, and then clicking the last file in the range. Multiple noncontiguous files can be selected by holding down the Command/Control key depending on the OS and selecting the noncontiguous files to upload.) For the tutorial, navigate to the location that the six images are saved. Select all six files to upload. Click "open."



Figure 2. Added files displayed in File List, boxed in red. The scrollbar is moved to the rightmost position.

To adjust how the file names are displayed in the msaGUI file list use **View > File Path** in the menu bar. Select **View File Only** to display just the file name. Select **View Parent** to display the file name and immediate parent folder. Select **View Full Path** to display the full path. For the file path "C:\Users\Name\Data\DataFile_1", the different display names are:

- View File Only: "DataFile_1"
- View Parent: "Data\DataFile_1"

- View Full Path: "C:\Users\Name\Data\DataFile_1"

To add additional data, click the "Add" button on the msaGUI. To delete data, click on the filename(s) to delete and then click the "Delete" button on the msaGUI.

2.3 Viewing Raw Data

Once data files are loaded, click on the filenames in the msaGUI to view the images. If multiple filenames are clicked, the first image clicked will be displayed. The file name and statistics about the image are displayed below the image. Mathematical representations for statistics are provided in **4.1. Statistics**.

At this point in the tutorial, if **View > File Path > View File Only** is selected and DataFile_24hr_1655 is clicked, the GUI should look like this:

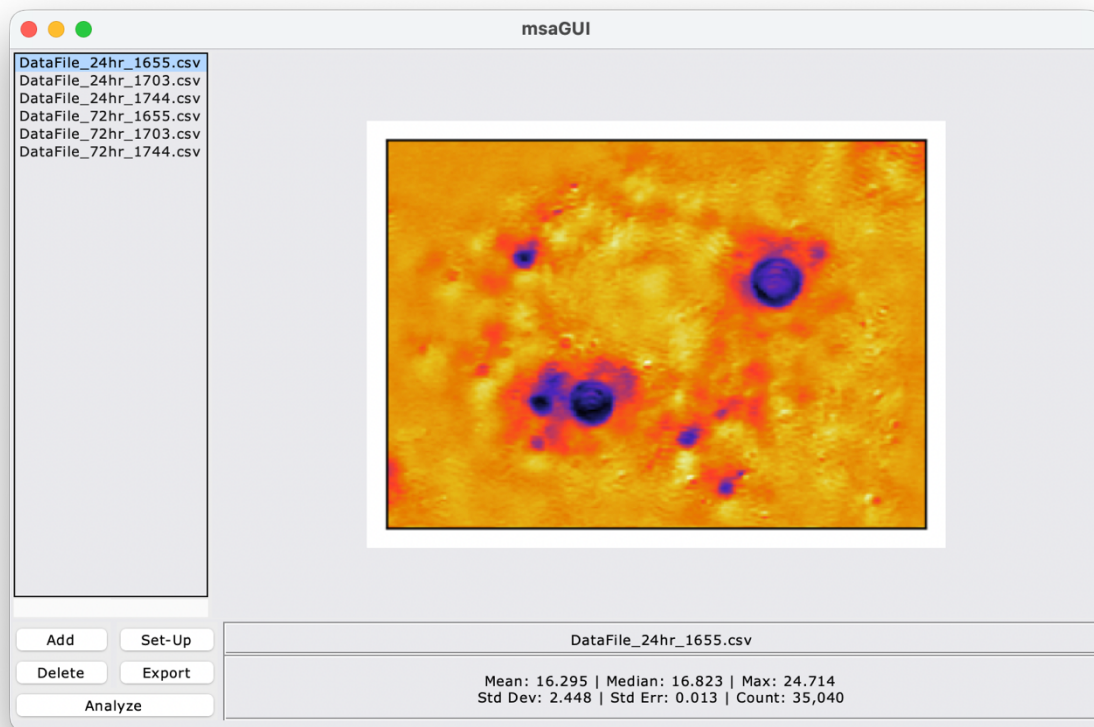


Figure 3. Selected file to display

To adjust the viewing area, drag the middle of the two panes or change the window size. After adjusting the window, reselect the file to implement the change.

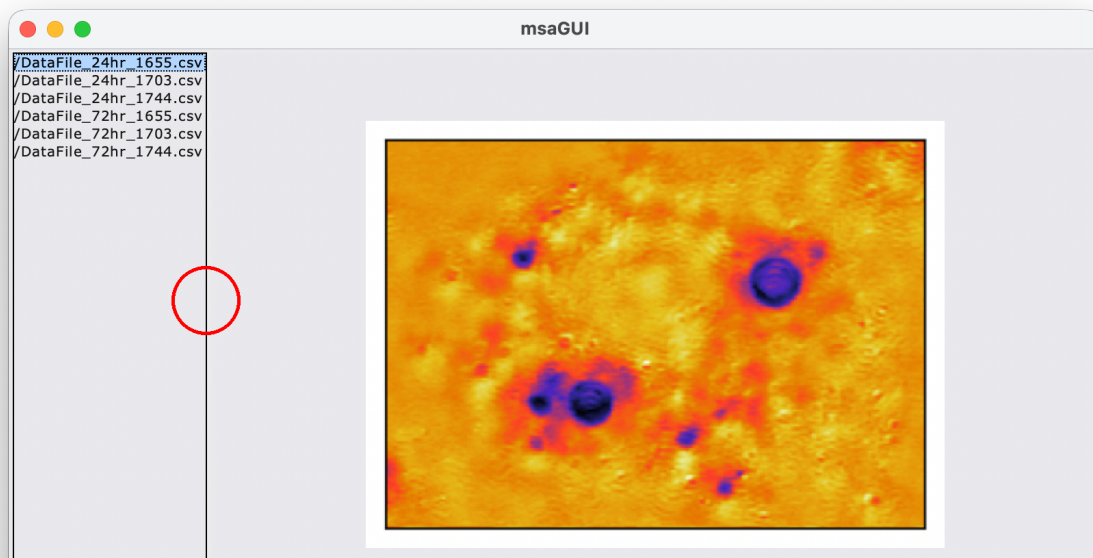


Figure 4. Red circle marks part of divider. Drag left or right to change relative size of file list and image

Other viewing options are available in the file menu **View**.

To view images as a histogram of pixel values, select **View > Histogram** from the menu bar or press Ctrl + H. Pressing Ctrl + H again will toggle back to an image view.

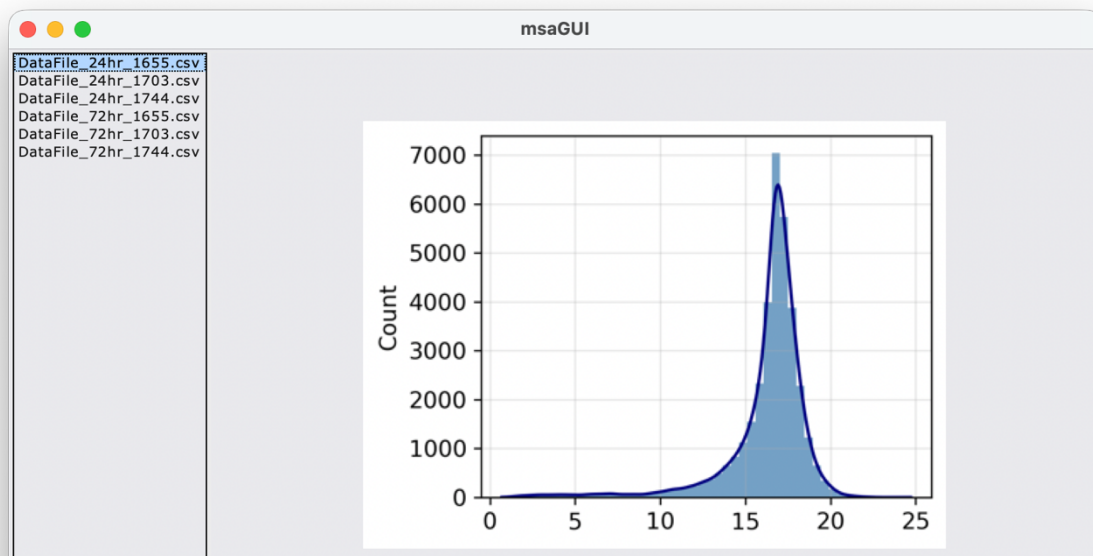


Figure 5. Histogram view of first data file.

2.4 Analysis Setup and Operations

Before analysis, it is important to define the image analysis pipeline. To do so, press the “Set-Up” button, which will pop up the Analysis Set-Up menu.



Figure 6. Button panel including the Set-Up button in the top right.

In the menu, define the seven steps shown in Figure 7. Each step is made up of four elements: a keyword, an operation, and output key, and either a second keyword or a value. Keywords are the text in file names that distinguishes between different types of images. The keywords used in this example are 1655, 1703, and 1744, which come from the file names.

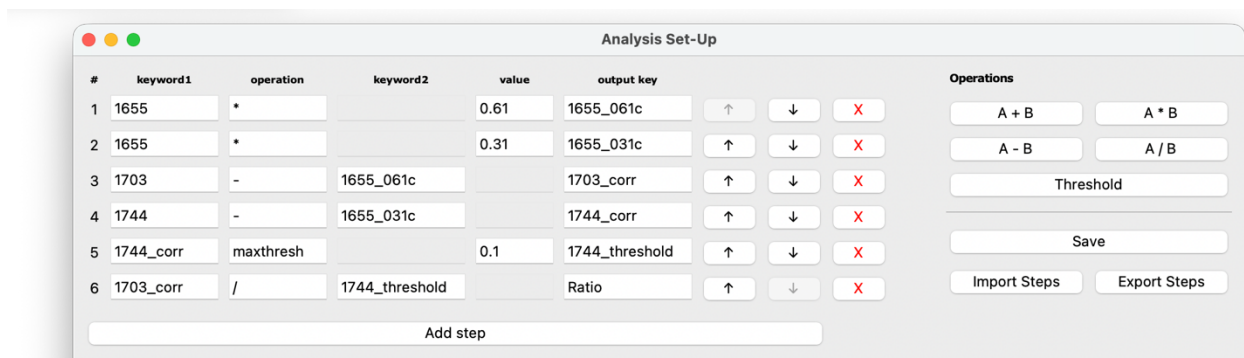


Figure 7. Panel for setting up controls. For each step includes keyword1, an operation (+, -, *, /, threshold, maxthresh), an output key, and either keyword2 or value depending on if the operation will occur to an image or not. Only either keyword2 or value can be selected at a time; the unselected option will be grayed out.

Steps can either be defined by manually typing in operations or by using the operations menu on the right side. In this example, you could create step 1 by pressing the $A * B$ button, which would pop up a dialog box to input the desired operation (Figure 8). However, for your convenience, **tutorial_steps.csv** is provided, which defines all the steps. Either input each step or import the steps via the “Import Steps” button. After, click **Save**. Pressing the window close button will cancel any changes.

Note that if desired, steps can also be exported for reuse. You can also rearrange or delete steps with the arrow buttons and X buttons.

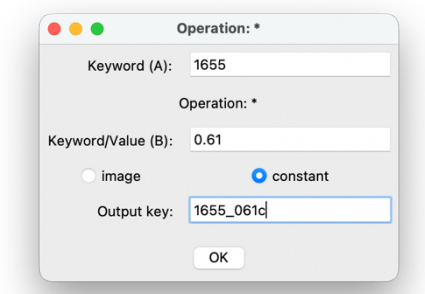


Figure 8. Multiplication operation button pop-up inputs to yield step 1. 1655, 0.61, and 1655_061c are keyword 1, value, and output key respectively. Constant, rather than image, is selected to indicate B as the value.

2.4.1. Mathematical Explanation of Operations

Mathematically, step 1 would create an image with keyword *1655_061c* with intensity (I_{1655_061c}), where

$$I_{1655_061c} = 0.61 \cdot I_{1655}$$

$I_{1655} =$	0	1	1
	1	1	1
	0	1	0

$I_{1655_061c} =$	0	0.61	1.22
	0.61	0.61	0.61
	0	0.61	0

Figure 9. The intensities of an example *1655* image and the resulting *1655_061c* image

Step 3 subtracts the *1703* image from the computed *1655_061c* image pixel-wise, i.e.,

$$I_{1703_corr} = I_{1703} - I_{1655_061c}$$

$I_{1703} =$	1	2	2
	1	2	2
	1	1	1

$I_{1703_corr} =$	1	1.39	0.78
	0.39	1.39	1.39
	1	0.39	1

Figure 10. The intensities of an example *1703* image and the resulting *1703_corr* image

Addition (+), subtraction (-), multiplication (*), division (/) all work in the same pixel-wise manner. Note that, for division by zero, undefined quotients are set to not-a-number (NaN). The threshold operation can be used in four different modes. The first is with a value representing a proportion, which will set any pixels in the image to 0 if they are below the maximum intensity in the image multiplied by the threshold value. For example, if the maximum value of the image is 1.39 and the threshold value is set to 0.1, any values below 0.139 will be set to 0. Mathematically,

$$T(x, p) = \begin{cases} 0 & \text{if } I(x) < pI_{max} \\ I(x) & \text{if } I(x) \geq pI_{max} \end{cases}$$

where $T(x)$ is the thresholded intensity, $I(x)$ is the intensity at position x , I_{max} is the maximum intensity in the image, and p is the threshold value represented as a proportion. A similar threshold mode is constant value, where pI_{max} is replaced by a single value independent of I_{max} . This value can be set by hand or automatically if the value is set to 'auto'. Automatic thresholding is determined via Otsu's method. The last method uses a reference image as a pixel-wise threshold for the maximum control.

Note that keywords can appear multiple times in steps and keywords can have overlapping phrases (e.g. "1703" and "1703_corr"). Near-matches are ignored (e.g. a keyword of "1703" and file of "Data_1704" will not be considered a match).

The end result of all operations in the example shown in Figure 8 yields a background corrected ratio image (I_{ratio}), which is mathematically calculated:

$$I_{ratio} = \frac{I_{1703} - 0.61 \cdot I_{1655}}{T(I_{1744} - 0.31 \cdot I_{1655}, 0.15)}$$

where I_{1703} is the image at frequency 1703 cm^{-1} , 0.61 is the correction factor for I_{1703} at the specified frequency for background correction, and I_{1655} is the image at the specified frequency for background correction. Analogously, I_{1744} is the image at frequency 1744 cm^{-1} , 0.31 is the correction factor for I_{1744} at

the specified frequency for background correction, and I_{1655} is the image at the specified frequency for background correction. T is the threshold function with two parameters where 0.15 is the threshold proportion and second parameter.

2.4.2. Grouping and Analyzing Data

msaGUI will automatically recognize files to group together for analysis by their similar filenames after excluding any keywords used to specify the frequencies for correction. The groups' file names and parent folder must be the same, excluding these keywords. For additional details on file naming convention, refer to **2.1 Data Preparation**.

The tutorial data are labeled by the wavenumber the image was collected, e.g. "DataFile_24hr_1655" indicates that the image was collected at 1655 cm^{-1} . DataFile_24hr_1744, DataFile_24hr_1703, DataFile_24hr_1655 all share a common name except for the keywords specific by the user-defined frequencies for correction and thus would be grouped together. Similarly, DataFile_72hr_1655, DataFile_72hr_1703, and DataFile_72hr_1744 will be grouped for analysis.

To analyze data,

1. Enter the desired steps into msaGUI as demonstrated above (Figure 7).
2. Click the "Analyze" button on the msaGUI and wait for the loading bar to reach 100%.
3. After analysis, the ratio images will appear in the file list. For the tutorial, the file list should appear as in Figure 11. By default, the thresholded out values will appear as white. To change this to black (as in the image), go to **Config > Image > Bad Value Color**, write "black", scroll down and click Save. See Figure 16 for more details.

2.5 Viewing Analyzed Data

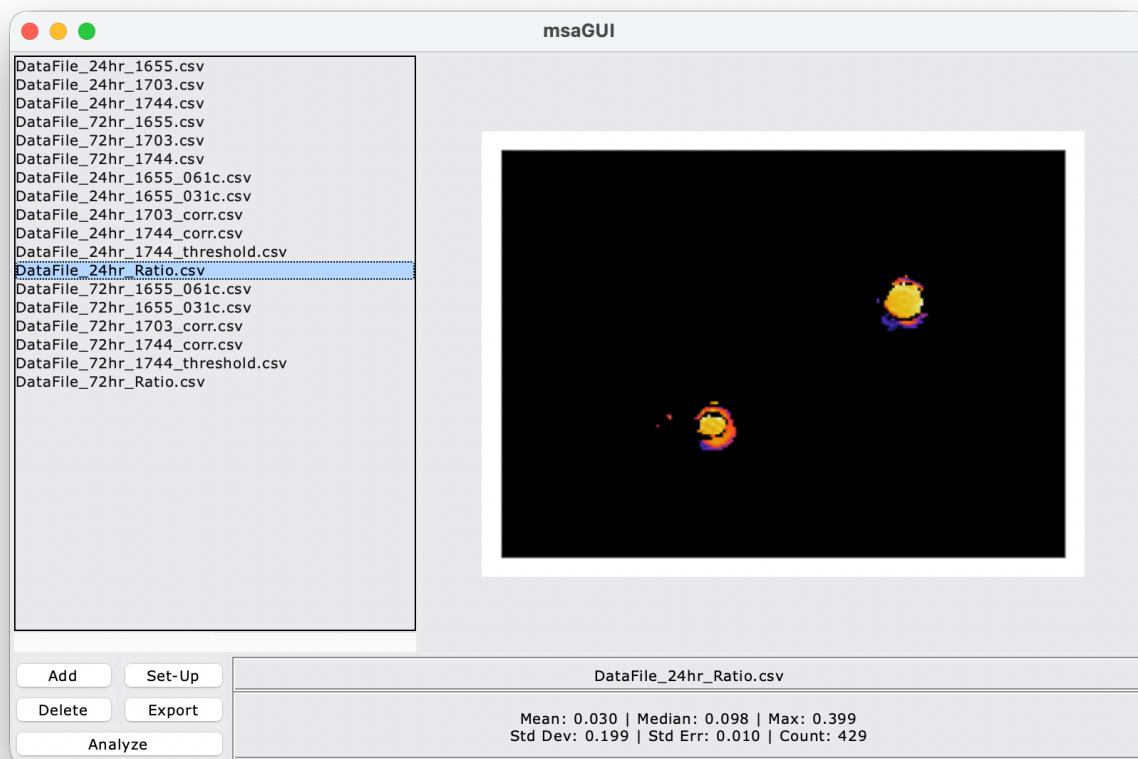


Figure 11. Processed image DataFile_24_Ratio

Now that the files are analyzed, the images are grouped and the output images are viewable. It is possible to filter for specific keywords under **View > Show**. Additionally, **View > Sort** provides five options for sorting: Time imported (into the program), Basename (i.e. the file name), Parent Path (i.e. the parent folder of the file), Group, and Keyword. To see all images from the same group in a single window, change the view to group view by clicking **View > Group View**. The images or group of images can also be displayed as a histogram **View > Histograms**. Note that loading groups will take longer than individual images.

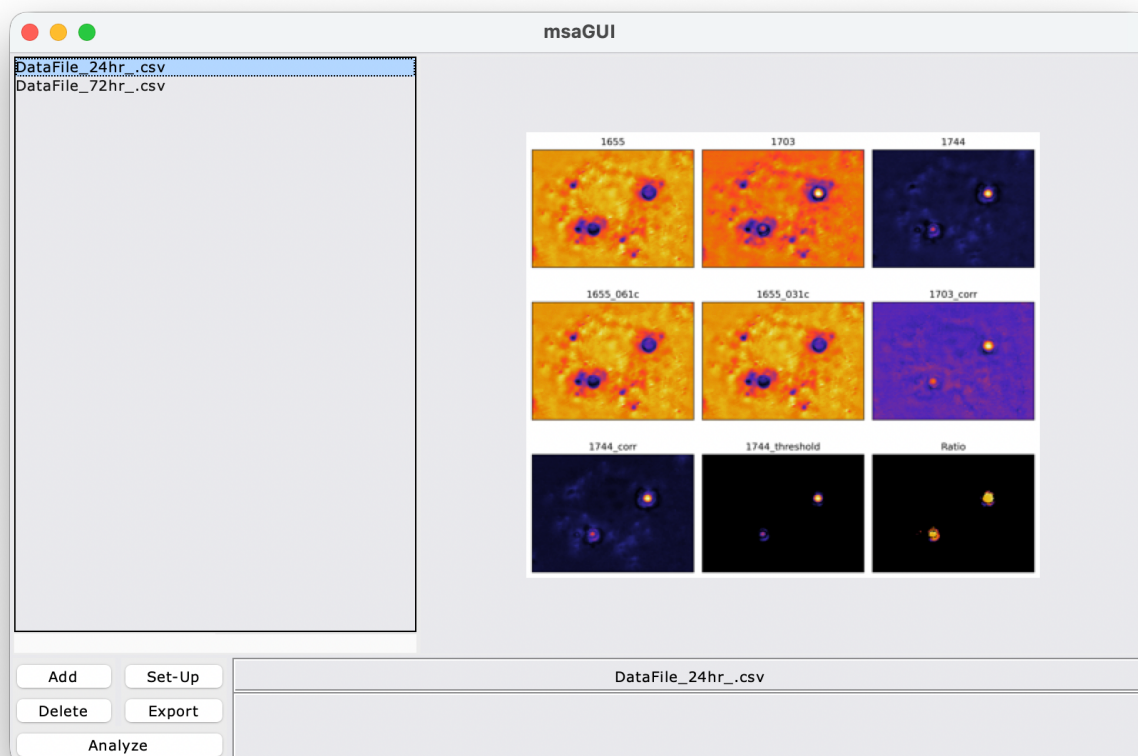


Figure 12. Group Image

2.6 Exporting Files and Statistics

Files are temporarily saved on the computer in the file msa_data.h5 in a temporary folder. Export images and corresponding statistics with the “Export” button.

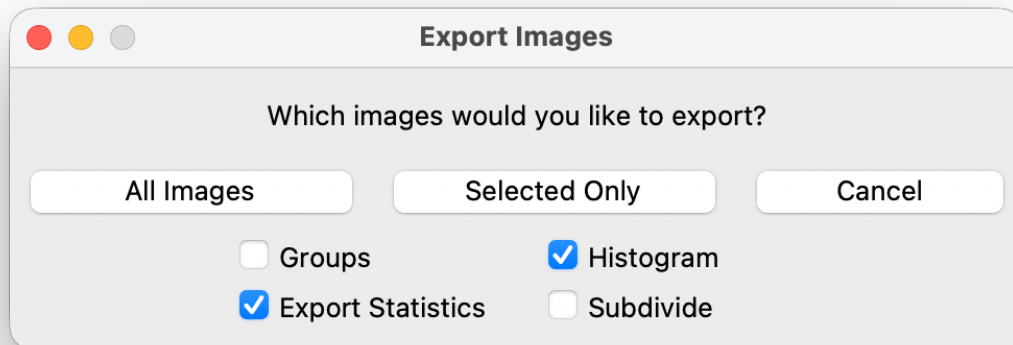


Figure 13. Export images panel.

Clicking All Images will export every image. Clicking Selected Only will only export images that are selected in the File List (See **2.6.1. Selecting and Rearranging Files**). Checking the Export Statistics box will also export the associated statistics with each image into a single CSV file called statistics.csv. Checking the Subdivide button will export files into subfolders named after the folder of their data origin. For example, the image from if an image is from “User/data/DataFile_24h_1655”, the exported path would be “{export_folder}/data/DataFile_24h_1655” rather than “{export_folder}/DataFile_24h_1655”. Checking Histogram will also export histograms. For this tutorial, check the Export Statistics and Histogram boxes and click All Images. After, select your desired folder for export. Files will be exported into the folder, as shown in Figure 14. Statistics are exported to statistics.csv (Figure 15).

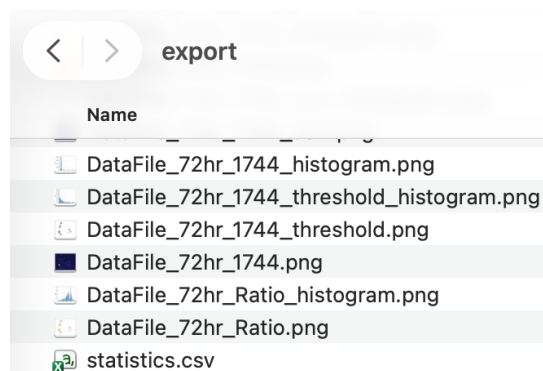


Figure 14. Export folder.

	A	B	C	D	E	F	G	H	I	J
1	nickname	group	keyword	kind	mean	median	max_signal	standard_deviation	standard_error	count
2	/export/DataFile_24hr_1655.csv	1	1655	input	16.29524126	16.82310499	24.71418	2.44763312164936	0.013075679673	35040
3	/export/DataFile_24hr_1703.csv	1	1703	input	9.691239641	9.999387	17.1229	1.528868066	0.008167478	35040
4	/export/DataFile_24hr_1744.csv	1	1744	input	5.414640244	5.049355999	76.14033	2.98678987218884	0.015955948330	35040
5	/export/DataFile_72hr_1655.csv	2	1655	input	15.62896539	16.72308	21.86112	3.14422894174123	0.018392965740	29223
6	/export/DataFile_72hr_1703.csv	2	1703	input	9.597507178	10.25759	20.47036	1.94783715384024	0.011394368127	29223
7	/export/DataFile_72hr_1744.csv	2	1744	input	5.596468853	5.128415	70.95621	3.47841735822821	0.020347885756	29223
8	/export/DataFile_24hr_1655_061c.csv	1	1655_061c	processed	9.940097168	10.2620941	15.07564979	1.49305620420611	0.0079796165	35040
9	/export/DataFile_24hr_1655_031c.csv	1	1655_031c	processed	5.051524790	5.215162550	7.661395799	0.75876626771130	0.004053461	35040
10	/export/DataFile_24hr_1703_corr.csv	1	1703_corr	processed	-0.2488575	-0.2650278	13.18844697	0.79666029570952	0.004255897	35040
11	/export/DataFile_24hr_1744_corr.csv	1	1744_corr	processed	0.363115453	-0.1826247	73.9118718	3.29473121828832	0.017601024287	35040
12	/export/DataFile_24hr_1744_threshold.csv	1	1744_threshold	processed	22.6298609	16.1191171	73.9118718	16.5850015474142	0.800731420197	429
13	/export/DataFile_24hr_Ratio.csv	1	Ratio	processed	0.029935491	0.0982443	0.399284481	0.19913472294626	0.009614315	429
14	/export/DataFile_72hr_1655_061c.csv	2	1655_061c	processed	9.533668893	10.2010788	13.3352832	1.91797965446215	0.011219709101	29223
15	/export/DataFile_72hr_1655_031c.csv	2	1655_031c	processed	4.844979273	5.1841548	6.776947199	0.97471097193978	0.005701819	29223
16	/export/DataFile_72hr_1703_corr.csv	2	1703_corr	processed	0.06383829	0.030396599	14.85363982	0.89385110848902	0.005228809	29223
17	/export/DataFile_72hr_1744_corr.csv	2	1744_corr	processed	0.751489579	-0.051392	68.0239882	3.93638419681903	0.023026879090	29223
18	/export/DataFile_72hr_1744_threshold.csv	2	1744_threshold	processed	19.11424451	13.9129255	68.0239882	13.6462599266611	0.492737887518	767
19	/export/DataFile_72hr_Ratio.csv	2	Ratio	processed	0.06539717	0.08188631	0.439381758	0.14985326481527	0.005410888	767

Figure 15. Exported statistics. The columns are nickname, group, keyword, kind, and the computed statistics. Nickname is the file path for input files or pseudo file paths for computed files. Pseudo file paths are placeholders to organize the computer files but are not where the computed files are stored. Keyword is the keyword attached for each file. Kind denotes if the image was provided by the user (i.e. input) or computed (i.e. processed). Count describes the total number of pixels.

2.6.1 Selecting and Rearranging Files

To select all files, click on one file in the list and press **Ctrl + A**. Alternatively, left click the top file and then, while holding down **Shift**, left click the bottom file. To select a subset of files, while holding down **Ctrl**, select the desired files. The selected files should be highlighted blue. You can click and drag individual files and groups of files around in the file list to change the order. This affects the order of the file list and the order that images are displayed in group view. Note: you cannot rearrange files in Group View.

3. ADDITIONAL FEATURES

3.1 Importing/Exporting File Lists, Settings, and Sessions

To save file lists for later, go to **File > Export > File List**. Import them with **File > Import > File List**. To export Settings, go to **File > Export > Settings**. To import them, go to **File > Import > Settings**. To set defaults for msaGUI, click **File > Export > Default Settings**. To save an entire session (i.e. settings, set-up, and files), **File > Save Session As...** will store the session as an H5 file, which can be loaded in with **Load Session**.

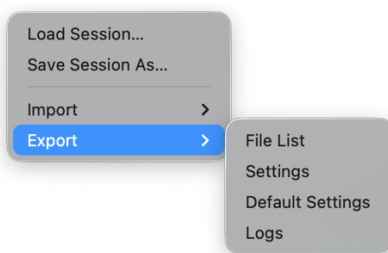


Figure 14. Import file menu. Options for import are File List and Settings

3.2 General Config

Several settings related to the data are customizable, to view the settings go to **Config**. Under **Config**, there are three setting types: **General**, **Image**, and **Histogram**.

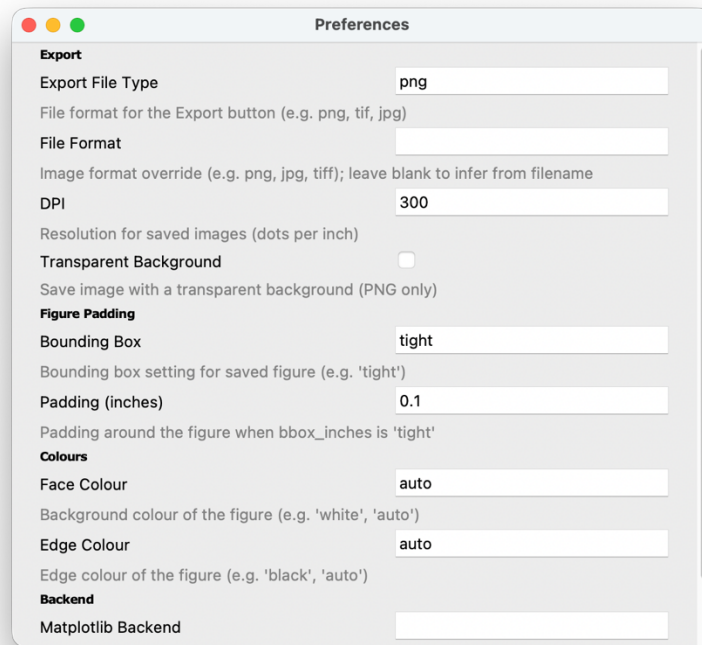


Figure 15. General Config menu. A scrollbar is provided on the right side.

3.2.1 Export

Export File Type: The file type that images will be exported as. A list of supported file formats is available at: <https://pillow.readthedocs.io/en/stable/handbook/image-file-formats.html>

DPI: Dots per inch. Defines the pixel density of the image.

Transparent Background: If checked, makes image have a transparent background. Only works for PNG file types.

3.2.2. Figure Padding

Padding (inches): The amount of padding (i.e. whitespace) around a figure.

3.2.3. Colors

Face Color: Background color of the figure. Default is white

Edge Color: Color of edge. Default is black.

3.3 Image Config

To yield publication-quality figures, outputted images are customizable (Figure 16). Customizable settings can be found under **File > Image Config**.

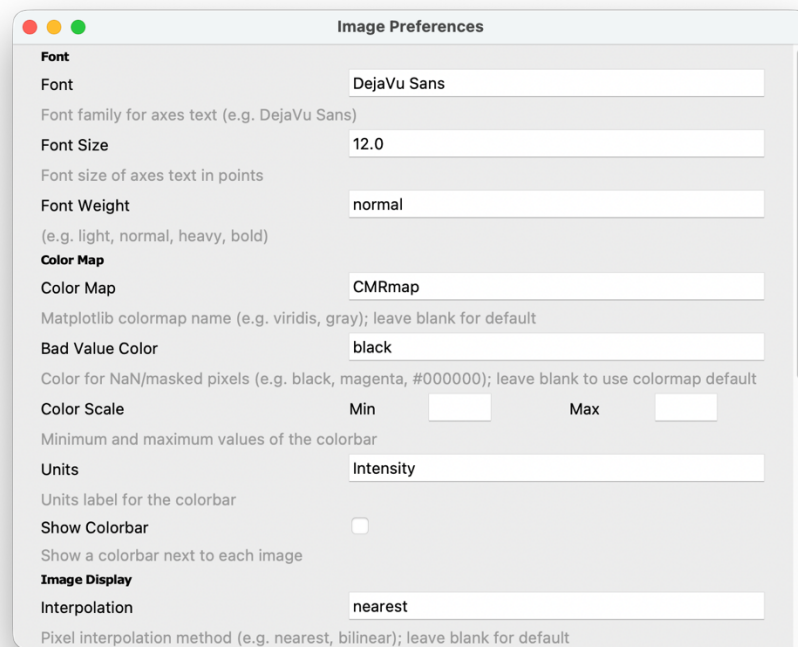


Figure 16. Image Config menu.

3.3.1 Font

The user can select the Font, Font Size, and Font Weight.

Font: Can be set to any font available on your computer

Font Size: The size of the text.

Font Weight: Style of font. See a full list available at:
https://matplotlib.org/stable/api/text_api.html#matplotlib.text.Text.set_fontweight

3.3.2 Color Bar

The color bar has three components: the color map, the scale, and the units.

Color Map: The palette used to indicate values. See a full list of available color maps here:
<https://matplotlib.org/stable/users/explain/colors/colormaps.html>.

Color Scale: The minimum and maximum values for the color range in images.

Units: The label put next to the color bar to indicate the axis. Can be any text. For example, units (e.g. "mV") or full label (e.g. "Intensity (mV)").

Show Colorbar: Toggle to show the colorbar.

3.3.3. Image Display

Interpolation: Method for scaling up images.

Origin: Defines origin for image. "lower" puts image origin in bottom left. "upper" puts image origin in top left.

3.3.3 Scale

Pixel Scale: Conversion between pixels and physical unit of measurement. Setting the pixel scale calibrates the scale bar fixed value (see below).

Scale Bar Units: Label of units

Scale Bar Color: Color of scale bar and label. See a full list of available colors:

https://matplotlib.org/stable/gallery/color/named_colors.html.

Scale Bar Location: Position on image where the scale bar is. Valid entries are combinations of one item from each of the following lists: (upper, center, lower) and (left, center, right). For example, "upper left" or "lower center". To put a scalebar in the center, simply write "center" (as opposed to "center center").

Scale Bar Fixed Value: The length (in physical units) of the scalebar. For no scale bar, set this value to 0.

3.3.4 Extra

Number of Tick Marks: Number of tick marks on both the x and y axes. For no tick marks, set to 0.

3.4. Histogram Config

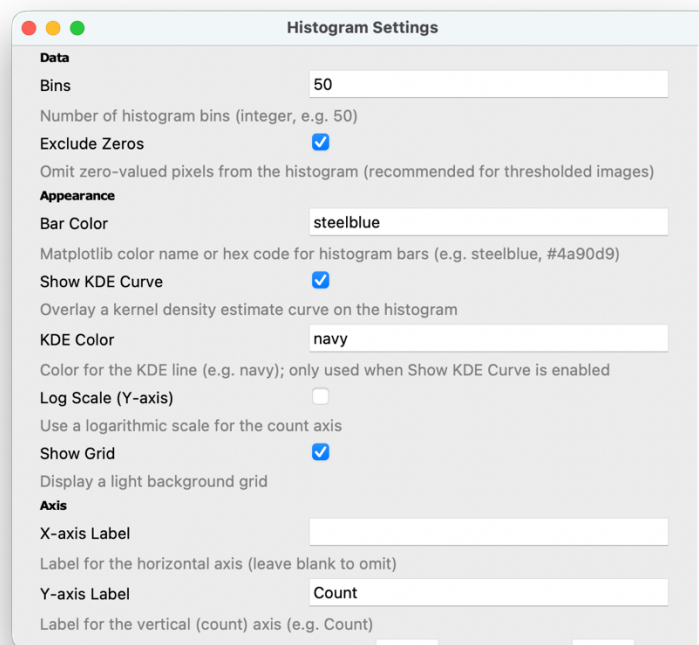


Figure 17. Histogram Config menu.

3.4.1. Data

Bins: Number of bins for the histogram

Exclude Zeros: Toggle to omit zero-values pixels from the histogram

3.4.2. Appearance

Bar Color: Color of the histogram bars. Can be named colors (see Scale Bar Color) or hex code (e.g. #4a90d9)

Show KDE Curve: Toggle for Kernel Density Estimation plot overlay.

KDE Color: Color of the KDE plot. Can be defined like Bar Color

Log Scale (Y-axis): Toggle for logarithmic scale.

Show Grid: Toggle for background grid.

3.4.3. Axis

X-axis Label: Label for X-axis

Y-axis Label: Label for Y-Axis

X-axis Range: Setting the displayed range of data. Leave blank for auto

3.4.4. Figure Size

Figure Size (per panel): Figure size (in inches) of each histogram plot. In group images, each plot is the specified size.

3.4.5. Font

Fonts are treated the same as in **3.3. Image Config.**

4. STATISTICS

Below are mathematical representations of the statistics displayed below images. For each, x is the value of each pixel.

Count (n): n is the number of pixels in an image.

Mean ($\bar{x}$), where:

$$\bar{x} = \frac{\sum x}{n}$$

Median ($\tilde{x}$), where For an ordered set: $x_1 \leq x_2 \leq \dots \leq x_n$:

$$\tilde{x} = \begin{cases} x_{\frac{n+1}{2}} & \text{if odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} & \text{if even} \end{cases}$$

Max (x_{max}):

$$x_{max} = \max(x_1, x_2, \dots, x_n)$$

Standard Deviation (Std Dev, σ):

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Standard Error (Std Error, SE):

$$SE = \frac{s}{\sqrt{n}}$$